BACKPROPAGATION

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* Model:

- $z = w \cdot x + b$
- $h = \tanh(z)$
- $\hat{y} = v \cdot h + c$
- $J = 1/2 (\hat{y} - y)^2$

Given: $x = 1$, $y = 2$, $w = 0.5$, $b = 0$, $v = 1$, $c = 0$, $\eta = 0.1$.

=) Forward Pass:

$z = w \cdot x + b = 0.5$

$h = 0.462117$

$\hat{y} = 0.462117$

$e = -1.537883$

$J = 1.182542$

=> Chain-rule derivations (symbolic)

local derivations:

- $\partial J / \partial \hat{y} = (\hat{y} - y) = e$
- $\partial \hat{y} / \partial c = \partial \hat{y} / \partial v \cdot h \partial \hat{y} / \partial h = \partial \hat{y} / \partial z \cdot w$
- $\partial h / \partial z = 1 - \tanh^2(z) = 1 - h^2$
- $\partial z / \partial w = x \partial z / \partial h = 1$

chain rule:

- $\partial J / \partial c = \partial J / \partial \hat{y} \cdot \partial \hat{y} / \partial c = e$
- $\partial J / \partial v = \partial J / \partial \hat{y} \cdot \partial \hat{y} / \partial v = e \cdot w$
- $\partial J / \partial b = \partial J / \partial \hat{y} \cdot \partial \hat{y} / \partial h \cdot \partial h / \partial z \cdot \partial z / \partial b = e \cdot w \cdot (1 - h^2)$
- $\partial J / \partial w = \partial J / \partial \hat{y} \cdot \partial \hat{y} / \partial h \cdot \partial h / \partial z \cdot \partial z / \partial w = e \cdot w \cdot (1 - h^2) \cdot x$